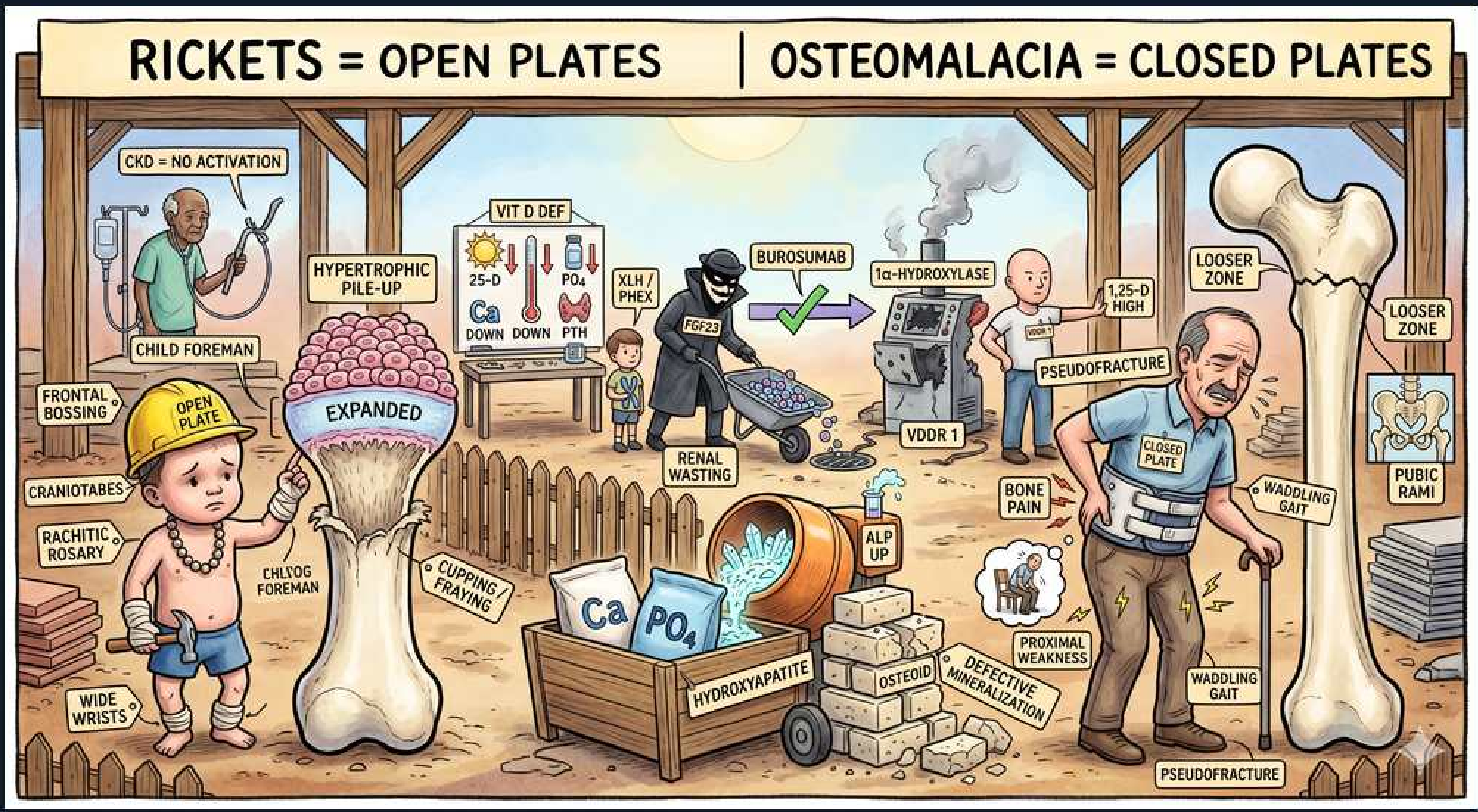
Calcium/PTH/VitD — The Mineralization Construction Site



1. Rickets = open plates banner

WHAT YOU SEE

Left header banner: RICKETS = OPEN PLATES — frames the whole left wing of the construction site as childhood disease where the growth plate is still open.

WHAT IT ENCODES

Rickets is defective mineralization of newly formed osteoid AND of the cartilaginous growth plate, occurring only while plates are still OPEN (children). Unmineralized cartilage piles up and is not resorbed, so the physis widens, cups, and frays on x-ray. The biochemistry is the same as osteomalacia (low/normal Ca, low PO4, high ALP, high PTH if vitamin-D driven); the only difference is an open vs closed physis. Most common cause worldwide is nutritional vitamin D deficiency.

BOARD TRAP

WRONG to call a widened, cupped, frayed metaphysis in a toddler 'physiologic growth' or scurvy — scurvy gives subperiosteal hemorrhage and a 'pencil-thin' cortex with normal ALP, whereas rickets shows a markedly ELEVATED ALP and a frayed, splayed physis. ALP separates them.

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2. Osteomalacia = closed plates banner

WHAT YOU SEE

Right header banner: OSTEOMALACIA = CLOSED PLATES — frames the right wing as adult disease where plates have fused, so the same defect shows up as soft, painful bone instead of deformed physes.

WHAT IT ENCODES

Osteomalacia is the identical mineralization defect but AFTER the growth plates have fused (adults), so there is no physis to deform — instead unmineralized osteoid accumulates on trabecular and cortical surfaces, producing soft, painful bone. Classic labs: low/normal Ca, low PO4, low 25-OH vitamin D, high ALP, high PTH (secondary). Hallmark x-ray finding is the Looser zone (pseudofracture). Treat the underlying cause: ergocalciferol/cholecalciferol for nutritional deficiency, calcitriol for CKD or 1-alpha-hydroxylase failure.

BOARD TRAP

WRONG to attribute an adult's diffuse bone pain + proximal weakness + low bone density to osteoPOROSIS. Osteoporosis has NORMAL Ca/PO4/ALP and is asymptomatic until fracture; osteomalacia has low PO4, HIGH ALP, low vitamin D, and Looser zones. Same skeleton picture, opposite labs — ALP and vitamin D discriminate.

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3. CKD = no activation

WHAT YOU SEE

Worker labeled 'CKD = no activation' unable to power the equipment.

WHAT IT ENCODES

The diseased kidney loses 1-alpha-hydroxylase, so 25-OH-D cannot be converted to active 1,25-(OH)2-D (calcitriol). The result is low calcitriol → low gut Ca absorption → hypocalcemia → secondary (then tertiary) hyperparathyroidism, plus phosphate retention because the failing kidney cannot excrete PO4. This is the core of CKD-mineral bone disorder (renal osteodystrophy). Treat with active vitamin D analogs (calcitriol, paricalcitol), phosphate binders, and calcimimetics (cinacalcet) — NOT plain vitamin D alone.

BOARD TRAP

WRONG to give plain ergocalciferol/cholecalciferol to a CKD patient with low 1,25-D and expect correction — the failing kidney cannot 1-alpha-hydroxylate it. You must give the ALREADY-ACTIVE form (calcitriol or paricalcitol). The discriminator is that 25-OH-D may be normal while 1,25-D is low.

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4. Frontal bossing / craniotabes

WHAT YOU SEE

The child laborer tagged with FRONTAL BOSSING, CRANIOTABES, RACHITIC ROSARY, WIDE WRISTS — each tag points at a soft, overgrown growth site where unmineralized osteoid deforms the bone.

WHAT IT ENCODES

These are the classic skeletal exam findings of rickets, all reflecting unmineralized osteoid at sites of rapid growth: frontal bossing (soft cranium remodels outward), craniotabes (ping-pong-ball softening of the occiput/parietal skull), rachitic rosary (beading at the costochondral junctions where cartilage overgrows), and widened wrists/ankles (metaphyseal flaring). Bowing of weight-bearing long bones (genu varum) appears once the child walks.

BOARD TRAP

WRONG to read a 'rachitic rosary' (rounded, symmetric costochondral beading) as the sharp-edged, tender 'scorbutic rosary' of scurvy — scurvy beads are angular with sternal displacement and accompany bleeding gums and perifollicular hemorrhage. Texture and associated signs separate them.

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5. Open plate / child foreman

WHAT YOU SEE

Child foreman with hard hat by an expanded/widened growth plate marked cupping and fraying.

WHAT IT ENCODES

At an open physis, chondrocytes proliferate and hypertrophy normally but the provisional calcification zone fails to mineralize, so cartilage is not removed and replaced by bone. The zone of hypertrophic cartilage piles up, widening the physis and producing the radiographic cupping, fraying, and splaying of the metaphysis. This is why rickets findings concentrate at the fastest-growing plates (distal radius, costochondral junctions, knees).

BOARD TRAP

WRONG to expect physeal changes in an ADULT with the same biochemistry — once plates fuse there is no physis, so the disease manifests as Looser zones/osteomalacia instead. Presence of an open plate is what defines rickets vs osteomalacia.

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6. Vit D def: Ca/PTH/PO4 down

WHAT YOU SEE

Central sign: vit D deficiency with Ca down, PO4 down, PTH up.

WHAT IT ENCODES

Nutritional/vitamin-D-deficiency rickets: low 25-OH-D → reduced gut Ca AND PO4 absorption → low-normal serum Ca → secondary hyperparathyroidism (high PTH) → PTH drives renal phosphate wasting, so phosphate falls further. Net pattern: low/low-normal Ca, LOW PO4, HIGH ALP, HIGH PTH, LOW 25-OH-D. Treat with high-dose vitamin D (e.g., ergocalciferol 50,000 IU/week ×6–8 weeks) plus calcium.

BOARD TRAP

WRONG to interpret the elevated PTH here as PRIMARY hyperparathyroidism. In primary HPT, Ca is HIGH; here Ca is low/normal, so the high PTH is APPROPRIATE secondary HPT defending calcium. The calcium level (low vs high) is the discriminator.

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7. Renal phosphate wasting / XLH-PHEX

WHAT YOU SEE

Saboteur with wheelbarrow labeled XLH / PHEX dumping phosphate, renal wasting.

WHAT IT ENCODES

X-linked hypophosphatemia (XLH) is the most common heritable rickets: PHEX gene mutation → excess intact FGF23 → renal phosphate wasting (low PO4) AND suppression of 1-alpha-hydroxylase, so calcitriol is inappropriately LOW/normal for the hypophosphatemia. Inheritance is X-linked dominant. Key labs: LOW PO4, high renal phosphate clearance (low TmP/GFR), normal Ca, normal/low 1,25-D, normal PTH, normal 25-OH-D. Same FGF23 mechanism drives tumor-induced osteomalacia.

BOARD TRAP

WRONG to treat XLH like nutritional rickets with vitamin D alone — the defect is renal PO4 wasting, not vitamin D deficiency (25-OH-D is normal). Plain phosphate + calcitriol is the old regimen; burosumab (anti-FGF23) is now preferred. The clue is LOW phosphate with NORMAL 25-OH-D and NORMAL calcium.

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8. Burosumab treatment

WHAT YOU SEE

A green CHECK MARK stamped over a BUROSUMAB label — the approval check marks the anti-FGF23 antibody as the modern fix specifically for FGF23-driven (XLH/TIO) phosphate wasting.

WHAT IT ENCODES

Burosumab is a fully human monoclonal antibody that binds and neutralizes FGF23, restoring renal phosphate reabsorption and calcitriol production. It is the preferred therapy for XLH and for tumor-induced osteomalacia when the tumor cannot be resected. It replaced the old burdensome regimen of oral phosphate + active vitamin D, which risked nephrocalcinosis and secondary/tertiary HPT.

BOARD TRAP

WRONG to give burosumab in vitamin-D-deficiency rickets or in any hypophosphatemia NOT driven by FGF23 — it only works when excess FGF23 is the cause (XLH, TIO, autosomal dominant hypophosphatemic rickets). Check that FGF23 is elevated and 25-OH-D is normal before reaching for it.

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9. 1-alpha-hydroxylase / VDDR1

WHAT YOU SEE

Machine labeled 1-alpha-hydroxylase converting; VDDR1 high 25-D, low 1,25-D.

WHAT IT ENCODES

Vitamin-D-dependent rickets type 1 (VDDR1) is an autosomal recessive loss of renal 1-alpha-hydroxylase (CYP27B1): substrate 25-OH-D accumulates (normal/high) but active 1,25-D is LOW. Result is hypocalcemia, low PO4, high ALP, high PTH. Treatment is physiologic-dose calcitriol (the active hormone), which bypasses the missing enzyme — contrast with VDDR type 2, an end-organ vitamin D receptor defect with HIGH 1,25-D and often alopecia, requiring high-dose calcitriol + calcium.

BOARD TRAP

WRONG to give plain vitamin D in VDDR1 expecting a response — the enzyme to activate it is absent, so 25-OH-D just keeps rising while 1,25-D stays low. Give calcitriol. The pattern HIGH 25-D + LOW 1,25-D is the tell for a 1-alpha-hydroxylase problem (VDDR1 or CKD).

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10. Hydroxyapatite cement / Ca+PO4

WHAT YOU SEE

Cement mixer pouring Ca and PO4 forming hydroxyapatite; osteoid and ALP up.

WHAT IT ENCODES

Mineralization is the deposition of hydroxyapatite [$Ca_{10}(PO_4)_6(OH)_2$] crystals onto the collagenous osteoid laid down by osteoblasts; it requires an adequate calcium-phosphate product and functioning alkaline phosphatase (ALP) to clear the inhibitor pyrophosphate. When mineralization fails, unmineralized osteoid accumulates and osteoblasts ramp up, so serum ALP RISES. Hypophosphatasia (loss of tissue-nonspecific ALP) is the mirror image — LOW ALP causes rickets because pyrophosphate cannot be cleared.

BOARD TRAP

WRONG to expect a LOW or normal ALP in active rickets/osteomalacia — ALP is characteristically ELEVATED. A normal ALP in suspected metabolic bone pain should redirect you toward osteoporosis (normal ALP) or, if ALP is paradoxically LOW, hypophosphatasia.

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11. Pseudofracture / Looser zones

WHAT YOU SEE

Adult marked with pseudofracture and Looser zones, bone pain.

WHAT IT ENCODES

Looser zones (pseudofractures, Milkman lines) are radiolucent bands of unmineralized osteoid running perpendicular to the cortex, typically at the medial femoral neck, pubic rami, scapula, and ribs. They are the pathognomonic radiographic sign of osteomalacia and represent incomplete stress fractures that fail to heal because new bone cannot mineralize.

BOARD TRAP

WRONG to call a Looser zone a true traumatic fracture or a Paget lytic lesion. Looser zones are symmetric, occur at characteristic sites, run perpendicular to the cortex, and lack a trauma history — they signal a global mineralization defect, not local injury. Bilaterality and location are the discriminators.

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12. Proximal weakness / waddling adult

WHAT YOU SEE

Adult with cane labeled closed plates, proximal weakness, waddling gait, pubic rami.

WHAT IT ENCODES

Adult osteomalacia presents with diffuse bone pain/tenderness, proximal MYOPATHY causing difficulty rising from a chair or climbing stairs, and a waddling (Trendelenburg) gait. The proximal weakness is partly from hypophosphatemia and low vitamin D effect on muscle. Insufficiency fractures (pubic rami, femoral neck) are common.

BOARD TRAP

WRONG to chalk up proximal weakness + waddling gait to polymyositis or a primary myopathy and order a muscle biopsy first — in this metabolic context check Ca, PO4, ALP, and 25-OH-D. Osteomalacic myopathy has bone pain, low PO4/vitamin D, and HIGH ALP; inflammatory myopathy has high CK and normal bone chemistry.

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13. Low PO4 alone = mineralization fails

WHAT YOU SEE

Bottom-right note: low PO4 alone -> mineralization fails.

WHAT IT ENCODES

Phosphate is half of the hydroxyapatite crystal, so isolated hypophosphatemia is SUFFICIENT to cause rickets/osteomalacia even with perfectly normal vitamin D and calcium. This is the unifying theme of XLH, tumor-induced osteomalacia, Fanconi syndrome, and chronic phosphate-binder/antacid overuse — all give rickets through phosphate depletion alone.

BOARD TRAP

WRONG to assume 'rickets/osteomalacia = vitamin D problem' and stop the workup at a normal 25-OH-D. If 25-OH-D is normal but PO4 is low, pursue a phosphate-wasting cause (FGF23 excess, Fanconi). The level of serum phosphate, not vitamin D, can be the whole story.